

Single core Copper & Aluminum Conductors

XLPE Insulated and PVC Sheathed

Voltage Grade

18 / 30 (36) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 19/33 (36) KV

MEDIUM VOLTAGE CABLES

Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating					Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground			Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Duct	Flat Touched	Trefoil		
mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	A	mm	kg/km	
A - Copper Conductors														
PM-CT14-X9-01-UP	50	0.3870	0.4937	0.4937	0.1436	0.5112	0.4649	226	218	189	246	241	31.0	1250
PM-CT15-X9-01-UP	70	0.2680	0.3420	0.3420	0.1615	0.4929	0.4367	278	267	232	308	301	32.8	1495
PM-CT16-X9-01-UP	95	0.1930	0.2465	0.2466	0.1762	0.4635	0.4173	331	316	277	373	365	34.5	1785
PM-CT17-X9-01-UP	120	0.1530	0.1957	0.1958	0.1898	0.4455	0.3993	375	362	317	429	420	35.9	2045
PM-CT18-X9-01-UP	150	0.1240	0.1588	0.1589	0.2042	0.4324	0.3862	415	404	353	484	474	37.6	2440
PM-CT19-X9-01-UP	185	0.0991	0.1273	0.1275	0.2215	0.4179	0.3717	466	456	402	556	545	39.4	2825
PM-CT20-X9-01-UP	240	0.0754	0.0974	0.0976	0.2434	0.4019	0.3557	535	528	468	655	642	41.9	3425
PM-CT21-X9-01-UP	300	0.0601	0.0783	0.0786	0.2643	0.3904	0.3441	594	591	527	748	735	44.3	4070
PM-CT22-X9-01-UP	400	0.0470	0.0621	0.0626	0.2898	0.3775	0.3312	663	665	600	852	842	47.0	5015
PM-CT23-X9-01-UP	500	0.0366	0.0496	0.0503	0.3209	0.3652	0.3190	742	749	682	979	969	50.5	6140
PM-CT24-X9-01-UP	630	0.0283	0.0398	0.0407	0.3594	0.3561	0.3099	833	842	755	1115	1108	54.8	7535
PM-CT25-X9-01-UP	800	0.0221	0.0328	0.0339	0.4006	0.3449	0.2987	910	931	863	1262	1257	59.4	9415
PM-CM26-X9-01-UP	1000	0.0176	0.0250	0.0253	0.4921	0.3438	0.2976	1071	1095	1026	1524	1529	70.2	12450
B - Aluminum Conductors														
PM-AT14-X9-01-UP	50	0.6410	0.8219	0.8219	0.1476	0.5040	0.4577	175	169	146	191	188	31.4	980
PM-AT15-X9-01-UP	70	0.4430	0.5682	0.5682	0.1624	0.4804	0.4342	214	207	179	239	234	32.9	1085
PM-AT16-X9-01-UP	95	0.3200	0.4106	0.4106	0.1781	0.4611	0.4149	256	247	216	290	284	34.7	1230
PM-AT17-X9-01-UP	120	0.2530	0.3247	0.3248	0.1907	0.4445	0.3982	291	281	246	334	326	36.0	1330
PM-AT18-X9-01-UP	150	0.2060	0.2645	0.2646	0.2071	0.4298	0.3836	323	315	276	377	370	37.9	1570
PM-AT19-X9-01-UP	185	0.1640	0.2109	0.2110	0.2244	0.4157	0.3695	365	358	316	435	428	39.7	1735
PM-AT20-X9-01-UP	240	0.1250	0.1611	0.1612	0.2434	0.4019	0.3557	421	414	366	514	503	41.9	1970
PM-AT21-X9-01-UP	300	0.1000	0.1292	0.1294	0.2643	0.3904	0.3441	466	465	414	590	577	44.3	2225
PM-AT22-X9-01-UP	400	0.0778	0.1010	0.1014	0.2898	0.3775	0.3312	533	530	477	682	670	47.0	2650
PM-AT23-X9-01-UP	500	0.0605	0.0794	0.0799	0.3209	0.3652	0.3190	602	605	550	779	781	50.5	3090
PM-AT24-X9-01-UP	630	0.0469	0.0625	0.0631	0.3604	0.3558	0.3096	686	686	621	920	907	54.9	3665
PM-AT25-X9-01-UP	800	0.0367	0.0502	0.0511	0.4006	0.3449	0.2987	775	775	719	1055	1042	59.4	4340
PM-AM26-X9-01-UP	1000	0.0291	0.0389	0.0391	0.4921	0.3438	0.2976	911	920	862	1290	1282	70.2	5820

The above data is approximate and subject to manufacturing tolerances